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07 March 2026
Total Citations: 2
Total Downloads: 1048

Published: 01 June 1992

Citation in BibTeX format

CHI92: Conference on Human Factors in Computing
May 3 - 7, 1992
California, Monterey, USA

Conference Sponsors:
SIGCHI

CHI '92: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (June 1992)
https://doi.org/10.1145/142750.142832
ISBN: 0897915135

TOWARD A MORE HUMANE KEYBOARD

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INTRODUCTION (W. HARGREAVES)

Occupational injuries related to computer use have become newsworthy. Most people who work in an office environment know a coworker who has experienced chronic pain or disabling injury to the upper extremities from computer use. Such injuries are commonly termed "repetitive strain injury" (RSI). To date, the focus for remedies has been on providing adjustable furniture and encouraging work breaks. Though the results of such changes have not yet been widely studied, informal inquiries in the workplace suggest that the problem remains and is growing.

The direct interface between most workers and their computers is the keyboard, and the belief that the traditional keyboard may be a major cause of computer-related RSI is gaining acceptance. Thus alternative keyboard designs are considered by some to be important factors in reversing or preventing RSI. A number of redesigned "ergonomic keyboards" are currently under development. Of these, three styles predominate: articulated traditional keyboards, chording keyboards, and hand-contoured keyboards.

Articulated traditional keyboards can exactly replicate a traditional computer keyboard, but they have at least one joint or hinge to allow adjustment of the right and left halves to each user's preference. Chording keyboards use key combinations to produce letters. These keyboards typically have only 8-10 keys, though each switch may have several active positions (e.g. ternary chord keyboard). The hand-contoured keyboard has keys arranged in separate right and left sections which are contoured to match the shape and movement of the hands and fingers. Typically the traditional Qwerty layout is retained in hand-contoured keyboards.

Significantly, there has been no forum for the side-by-side comparison of objectives, features, and demonstrated effectiveness for the different styles of ergonomic keyboards. The intent of this panel is to create such a forum. The first two panelists will present objectives and test methods. The last three will contribute specific attributes of articulated, chording, and hand-contoured ergonomic keyboards.

MEASURING KEYBOARD-RELATED MUSCULOSKELETAL STRAIN (D. REMPEL)

Repetitive strain injuries due to office work more than doubled from 1989 to 1990 [1] and are probably due to the increased time workers spend using a single tool, the computer keyboard. These injuries range from muscle fatigue to nerve entrapments and tendon-related disorders. Modifications to the current alphanumeric Sholes brothers' keyboard (Qwerty) have been proposed to make the keyboard safer, but few of these modifications have been rigorously tested to determine whether they meet stated claims. Force is considered a significant risk factor for these injuries, and reducing the keyswitch force of activation (light touch keyboard) has been shown to reduce the force that typists apply [5]. Other work-related risk factors are repetition of a specific motion, awkward joint postures (e.g. wrist in extension), and long durations of

exertion without adequate recovery time [6]. Engineering modifications to a keyboard can only alter these risk factors to a certain extent. Task modification and appropriate work-rest cycles will probably play a greater role in reducing these risk factors.

To determine the health impact of keyboard designs, tests can be either of short-term (minutes to several days) or long-term (months to years) duration. Although properly conducted long-term testing is more valuable, it as also more expensive. The most important elements in determining the value of either short or long-term tests are (a) study bias, (b) predictive value of the measurement tool, and (c) number of subjects tested.

Study bias is easily introduced, since it is almost impossible to blind subjects or researchers to keyboards of a different design and have them use the keyboards at the same time. Therefore, subjective measures (How does this keyboard feel?, Are your wrists sore after using this keyboard? Is your typing speed better with this keyboard?) are difficult to interpret. Objective measures are available (typing speed, error rate, joint angle, impact force, muscle electrical activity, nerve function, physical examination, intracarpal pressure) but require experience to collect and interpret. Each of these objective parameters measures a different physiologic aspect of typing and each has a different value in predicting repetitive strain injuries. Finally, if a study has too few subjects, the results may be meaningless.

DESIGN CRITERIA — WHAT SHOULD AN ERGONOMIC KEYBOARD DO, AND HOW CAN WE TELL IF IT DOES IT? (N. HALPERN)

The keyboard has become a common human-machine interface due to the widespread use of video display terminals (VDTs). The tasks involved in VDT operations are varied, yet most require a keyboard as the input device. Only a few studies have investigated the effects of keyboard design on the musculoskeletal system of the operators. Most studies looked at the key layout and emphasized the measurements and mechanical parameters of the keyswitches. In order to investigate the ergonomics of keyboard design we need to examine the demands on the human operator.

The demands of keyboard operation relate to *energy consumption* and *information processing*. Design criteria need to address energy consumption from the viewpoint of both localized dynamic muscular work and static postural effort. Information processing demands relate to visual identification and feedback. The design goals of an improved, ergonomic keyboard should be to:

(1) Optimize the work of muscle groups involved in keying.

(2) Optimize tactile feedback.

(3) Reduce static postural effort.

(4) Facilitate visual identification of keys.

The interest in RSI associated with VDT work indicates the need for a special focus on the muscular demands of this work. Static or postural effort in keyboard operation involves the muscles of the upper extremities, mainly to support the fingers; the dynamic component for these muscles may be small. Stress in the upper extremities is caused by a combination of forearm pronation, ulnar deviation of the wrist, extension of the hand to aid the reach of the smaller fingers, arm abduction, and shoulder elevation. An injury mechanism can be inferred from the few relevant studies.

The important design parameters that affect static postural effort are keyboard size and shape, wrist and arm support, and key layout. Dynamic work performed during keyboard use primarily involves the small muscles of the fingers. Design parameters that affect this effort are key travel, key shape, inter-key distance, keying frequency, keyboard shape, and key assignment (layout). The latter probably also affects the dynamic work performed by muscles in the wrists and forearms.

The presentation will outline ways to achieve ergonomic design goals, taking into consideration the influence of workstation design and work organization parameters on the human operator.

DESIGN LESSONS FROM TROUBLED UPPER LIMBS (R. MARKISON)

The human lifespan was 25 years in ancient Greece, and the present human lifespan is three to four times as long. The author's experience in dissecting thousands of troubled limbs over a 17-year period underscores a pressing need to design tools which work with, rather than against, the delicate form and function of the upper limbs.

As a design-minded hand surgeon, I would like to address a handful of issues:

(1) Flat, non-adjustable keyboards violate the golden rule of muscle function by using muscles at inappropriate fiber lengths.

(2) Non-adjustable keyboards abuse the thumb, which after all, is only 40,000 years old in the context of more than two million years of human hand evolution.

(3) Thirty-five percent of hands have congenital linkage of the ring and little finger flexor tendons; this has great design implications for keyboard development.

(4) Pronation of the forearm combined with ulnar deviation of the wrist and abduction of the little finger away from the hand cause significant difficulties with the ulnar and median nerves, as well as with the tendons that traverse nine separate discrete tunnels at the wrist.

(5) Adjustability is absolutely essential, since we have a broad age span of workers and a broad anthropometric variation among workers who may have very tiny hands or very large hands. Examples will be drawn from repetitive strain injuries among musicians [4].

UNCONVENTIONAL KEYBOARDS — FOR EXAMPLE, THE TERNARY CHORD KEYBOARD (K. KROEMER)

RSI, though known for decades to afflict keyboard operators, has now become an area of major concern. This is partly because of the widespread use of keyboards, and possibly because of heightened health awareness. While numerous proposals have been made to change the keyboard layout since the first commercial typewriter was developed about 125 years ago, the current computer keyboard layout is in essence unchanged from the original. Standardized in 1966, the traditional keyboard has features that demand improvements for biomechanical and psychophysical reasons. These include the key *per se*, the large number of keys, their layout on the keyboard, and the arrangement of the keyboard with respect to the operator.

There are several design variables for keys and keyboards currently used, or proposed. These can be categorized by (1) keyswitch parameters (2) position of the keyboard, and (3) data entry mode (see Table 1). Many of the previously proposed alterations were based on theoretical considerations, some of which are still valid (e.g., number of keys, layout, posture) while others are not (e.g. letter-key-digit allocation).

Each of these variables, alone or combined, affect the user's digit posture, wrist posture, arm posture, and movements (and thus energy requirements). These variables, together with the temporary task requirements, determine the person's musculo-skeletal strain (risk of RSI) and performance (number of entries, errors.).

A comparison of existing or proposed designs indicates that, depending on their specific features, they might affect keyboard operators' health and performance in various ways. We performed experiments on specially designed apparatus to measure finger mobility, strength, and speed. We also had subjects perform extensive learning and performance trials on a prototype of a ternary chord keyboard. The comparison of biomechanical digit characteristics and keying performances showed no or little correlation, which is quite in contrast to previous presumptions.

The results of our survey and experiments have implications for the design, arrangement, and use of future keyboards.

HAND-CONTOURED KEYBOARDS (J. LITEWKA)

It was recognized early on that the Sholes/Qwerty keyboard required awkward postures. Over the last 100 years, many concepts and experimental designs have been introduced to remedy this deficiency. Most early improvements were intended to promote faster typing. However, with recent increases in both incidence and awareness of RSI, keyboard safety has become a paramount issue without detracting from the importance of productivity. As a result, there is a new objective for keyboard designers: minimize the stressful postures, force, and repetition which combine to cause keyboard-related injuries.

In the mid-1970s, I participated in initial research and development of the first commercial hand-contoured keyboard [reviewed in 2]. This keyboard, in limited commercial use for ten years, pioneered ergonomic

TABLE 1. KEYBOARD VARIABLES (KROEMER)

Keyswitch Parameters	Keyboard Attitude	Data Entry Mode
Activation mode	Slope	Activation mode
Activation energy	Slant	(displacement, impulse)
Number of states per key	Tilt	Single (traditional)
Number of keys		Multiple (chording)
Key arrangement		

features intended to reduce strain while maintaining or increasing typing speed and accuracy. Its main features are concave wells containing keys for the right and left hands, separated by a central space. Additional sets of keys are operated by the right and left thumbs. The use of these thumb keys redistributes the work load among the digits, reducing the work done by the little fingers and minimizing ulnar deviation which would otherwise occur when reaching for the outermost keys. Overall geometry reduces pronation, ulnar deviation, wrist extension, and little finger abduction, while providing shorter, simpler reaches for all fingers. Long term use by injured users supports the efficacy of this design in reducing or eliminating pain.

In 1991, an enhanced hand-contoured keyboard with additional features and a reduced learning requirement was developed in the U.S. Location and accessibility of function keys and thumb keys were improved, as was keyswitch tactility. By incorporating built-in palm rests, static postural effort was reduced. The standard layout of the 101-type computer keyboard was largely maintained to minimize learning requirements. The product conforms to a broad range of hand sizes and shapes, from those of children to those of large adults, without the need for moving parts. Importantly, neither the British nor the U.S. designs can be misadjusted by a naive user.

A learning-curve study was conducted with healthy subjects in late 1991, using a prototype of the U.S. hand-contoured keyboard [3]. Additional one- to three-day evaluations by RSI-sufferers have been carried out. Most healthy subjects and all injured subjects preferred the hand-contoured design over the traditional one, finding the former more comfortable and requiring less effort. In the learning-curve study, all subjects reported the hand-contoured keyboard easy to learn, with the majority typing faster than on a traditional keyboard (with equivalent accuracy) after less than eight hours. Injured subjects uniformly reported significant relief from symptoms and increased output.

It appears that an optimized hand-contoured ergonomic keyboard can reduce RSI risk factors without sacrificing productivity or imposing a significant learning curve.

REFERENCES

1. BLS Reports on Survey of Occupational Injuries and Illnesses in 1977-1990. Washington, D.C.: Bureau of Labor Statistics, U.S. Dept of Labor, 1991.

2. Hobday, S.W.. A keyboard to increase productivity and reduce postural stress. In Trends in Ergonomics, Human Factors V. F. Aghazadeh, ed., Elsevier Science Publishers B.V., North Holland, 1988, pp. 321-330.

3. Jahns, D.W., Litewka, J., Lunde, S.A., Farrand, W.P., and Hargreaves, W.R. Learning curve and performance analysis for the Kinesis™ ergonomic keyboard — a pilot study. Presented as a poster at the HFS 35th Annual Meeting (San Francisco, Calif., Sep 2-6, 1991). Copies available from Kinesis[a].

4. Markison, R.E. Treatment of musical hands: redesign of the interface. In Hand Injuries in Sports and Performing Arts. Amadio P.C., Hand Clinics 6,3 (Aug 1990), 525-544.

5. Rempel, D., Gerson, J., Armstrong, T., Foulk, J., Martin, B. Fingertip forces while using three different keyboards. In Proceedings of the HFS 35th annual meeting (San Francisco, Calif., Sep 2-6, 1991). Human Factors Society, Santa Monica, Calif., pp. 253-255.

6. Rempel, D., Harrison, R.J., Barnhart, S. Work-related cumulative trauma disorders of the upper extremity. JAMA 267,6 (Feb 12, 1992).